Predicting Rehospitalization of Diabetes Patients

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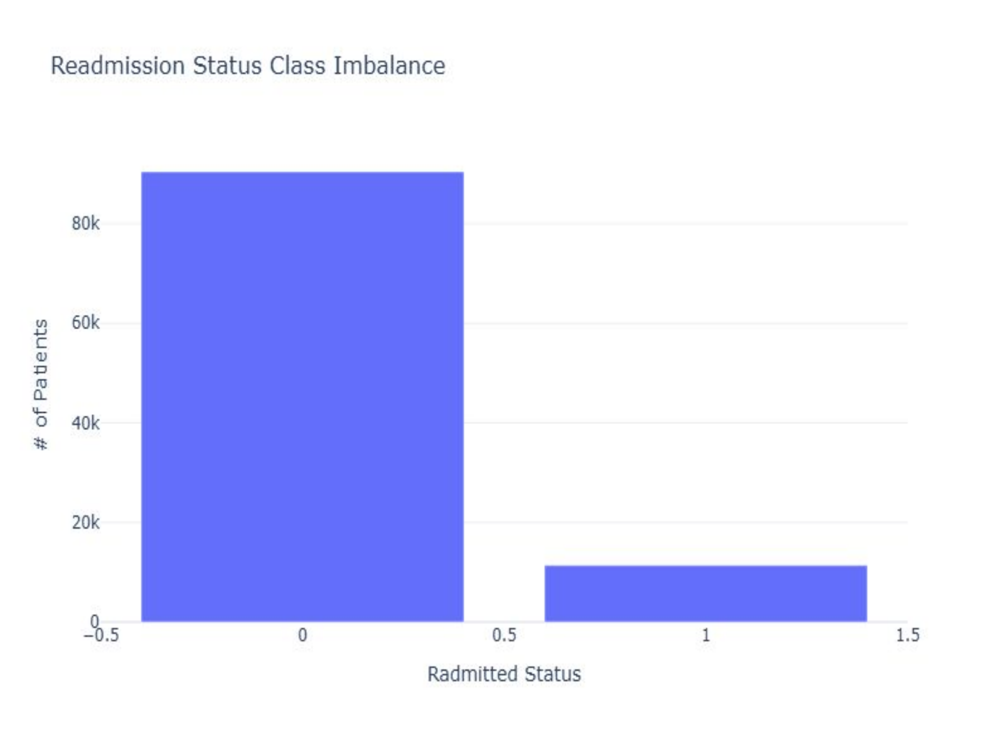
Background

- Diabetes 130-US Hospitals for Years 1999-2008
    - Acquired from UC Irvine Machine Learning Repository
- Shape:
    - Features: 50
    - Entries: 101,766
- Feature Preview:
    - General information
        - sex, age, race
    - Visit information
        - # of lab tests/procedures
    - Hospital information
        - # of visits, length of stay, # of emergency visits
    - Diabetes medication dosage (22 features)
        - ‘steady’, ‘up’, ‘down’, ‘no’

Objectives

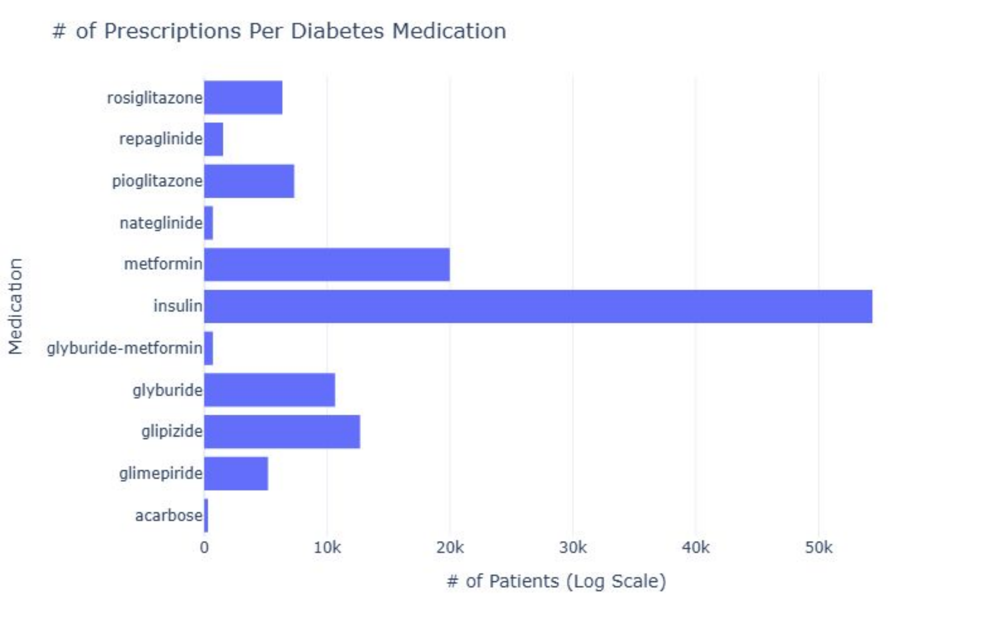
1. Classify “readmission” within 30 days of discharge
2. Improve clinical outcomes
3. Reduce hospital costs
4. Reduce mortality rates



EDA

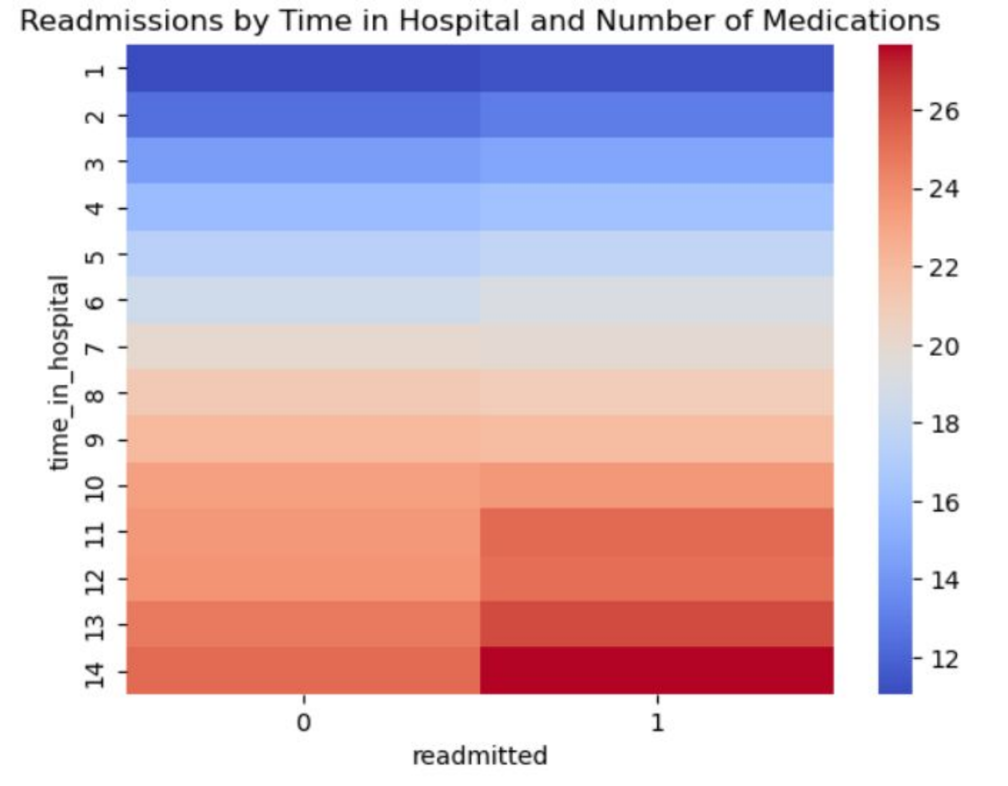
Medication Feature Imbalance

- High imbalance across medications
- Insulin and metformin are the most common
- Sparse features can reduce model performance
- Solutions: feature selection



Severity Feature Motivation

- Positive trend for time in hospital and number of medications for readmitted patients
- 10-14 days shows the strongest difference
- Potential feature engineering combination



Logistic Regression Feature Importance

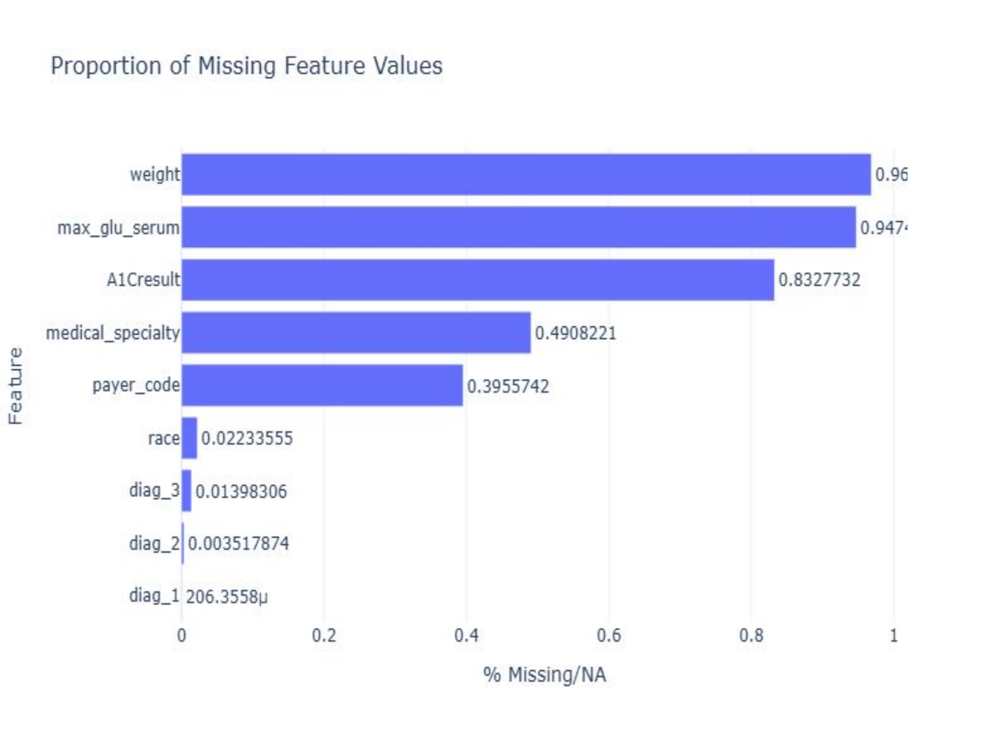
- Used a baseline logistic regression model and coefficient to compare importances
- medical _specialty appeared 6/20 times
- Potential cause: appeared in few observations
- Solution: grouping

Important Features:

	Feature	Importance
18	cat__medical_specialty_ObstetricsandGynecology	0.757197
33	cat__chlorpropamide_No	0.681380
48	cat__acarbose_Steady	0.647956
12	cat__age_[10-20)	0.633751
23	cat__medical_specialty_Surgery-Vascular	0.609547
27	cat__repaglinide_Down	0.607670
34	cat__chlorpropamide_Steady	0.601264
47	cat__rosiglitazone_Down	0.525172
26	cat__metformin_Up	0.458377
11	cat__age_[0-10)	0.423078
22	cat__medical_specialty_Surgery-Cardiovascular/...	0.403815
19	cat__medical_specialty_Orthopedics-Reconstructive	0.398335
46	cat__pioglitazone_Steady	0.374095
16	cat__medical_specialty_Cardiology	0.344427
42	cat__glyburide_Down	0.329273
4	num__number_inpatient	0.306732
24	cat__medical_specialty_Urology	0.298650
49	cat__acarbose_Up	0.295619
74	cat__diag_3g_neoplasms	0.294264
61	cat__glipizide-metformin_No	0.270914

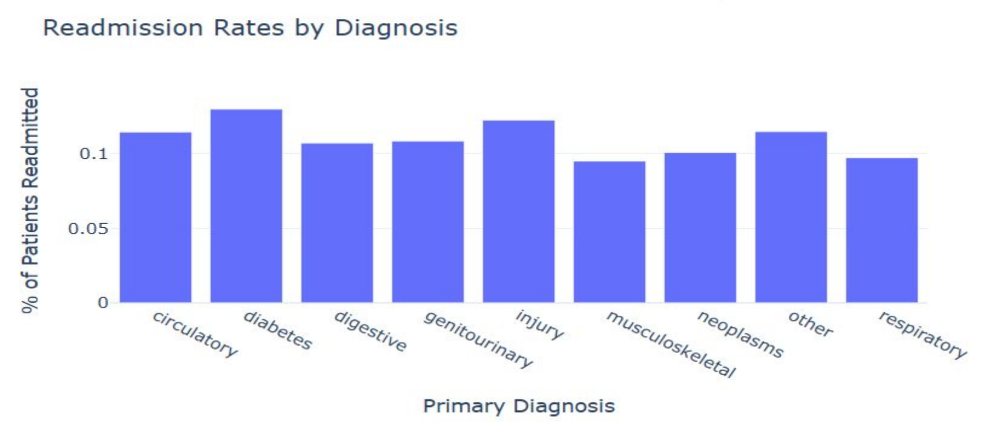
Missing Values

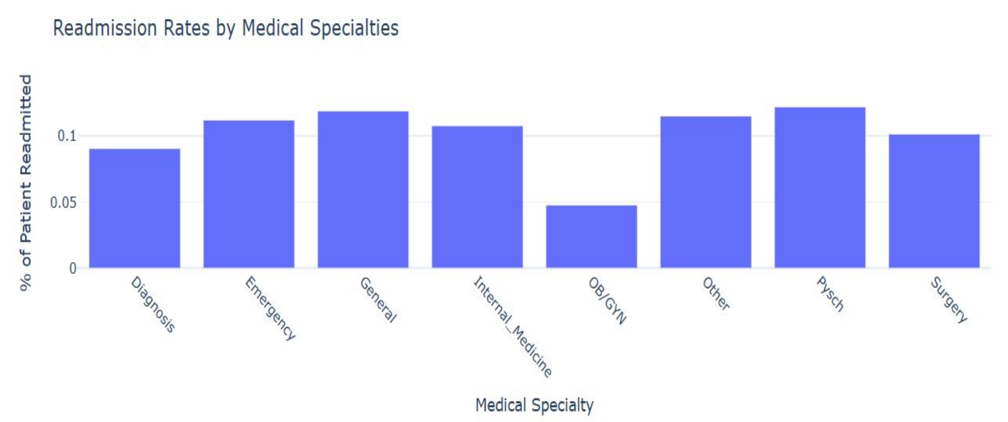
- Weight/blood test data missing for most patients
- Potentially feature engineer highly missing features to binary
    - Tests can reflect on the patient's condition
- medical _specialty can be affected by hospital documentation (lack of a controlled vocabulary)



Feature Engineering

- Severity
- Total_visits
- Grouping ICD9 Codes
  - 1000 codes -> 9 groups
- Grouping ‘medical_specialties’
  - 72 specialties -> 8 groups





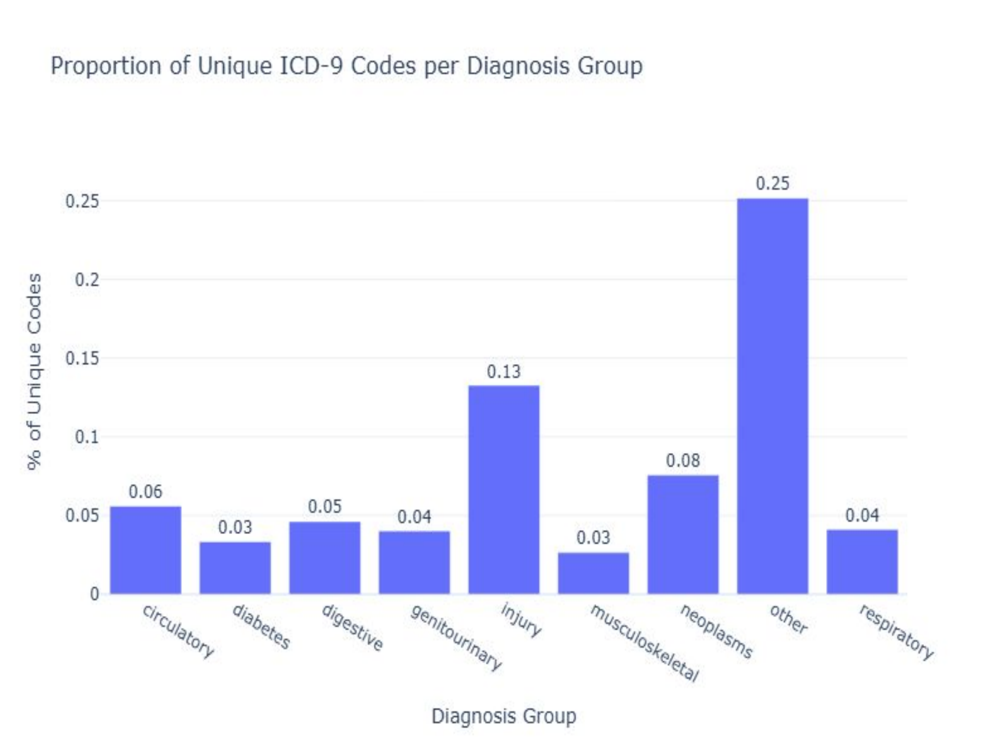
Dimensionality Reduction

Grouping ICD9 Codes

- Before grouping average:
  - 0.00139
- After grouping average:
  - 0.11111

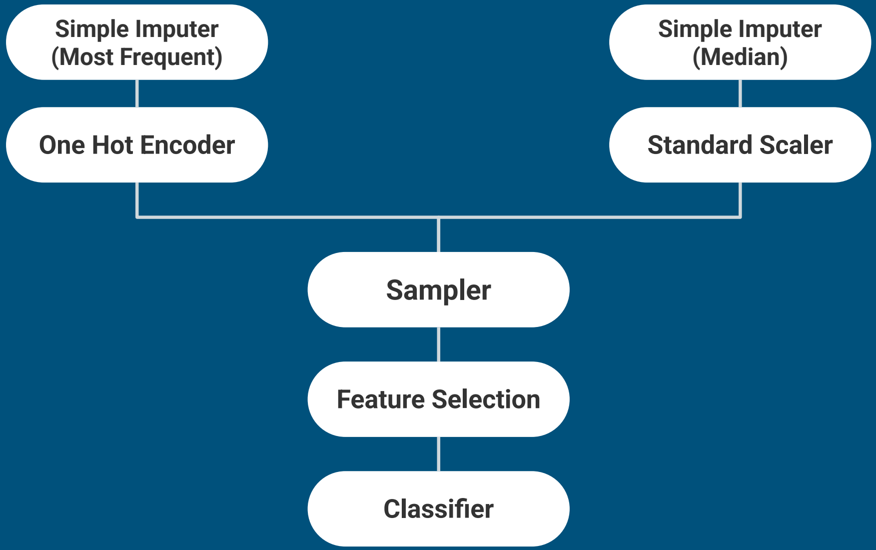
Grouping Medical Specialties:

- Before grouping average:
  - 0.01388
- After grouping average:
  - 0.125



Pipeline Components

- Preprocessing
  - OneHotEncoder
  - StandardScaler
- Imblearn Samplers
- Feature Selection:
  - Variance Threshold
  - Select K Best (+ Engineered)
  - Linear SVC selector



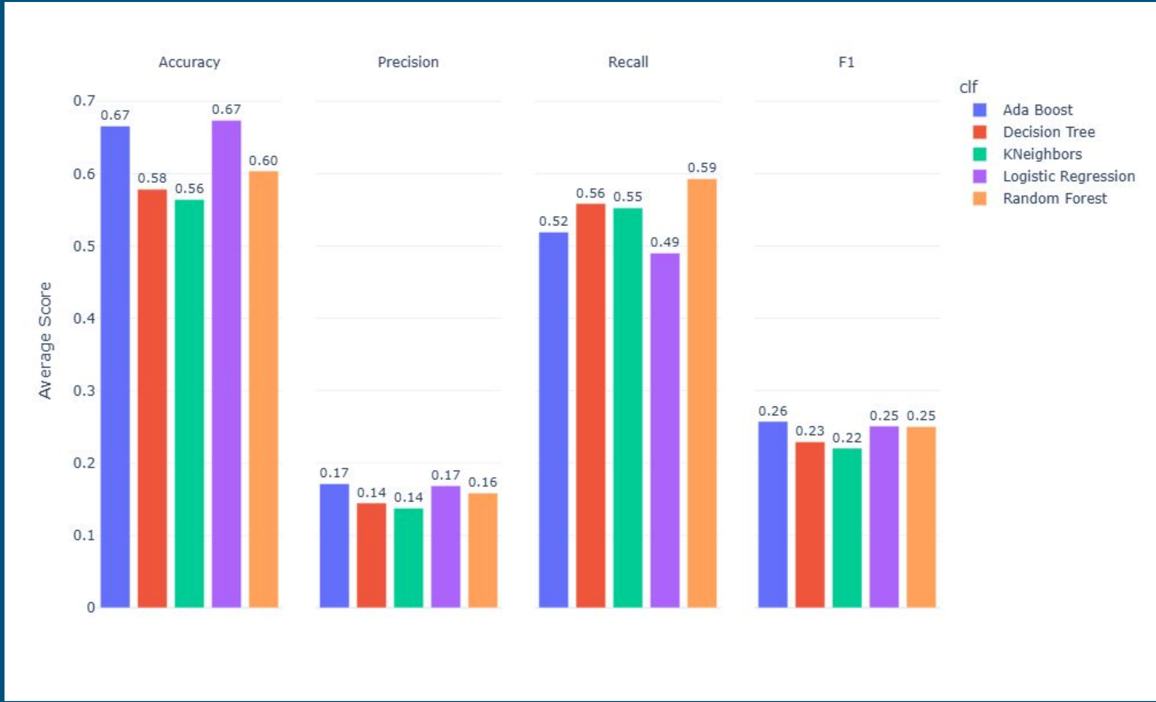
Models Used

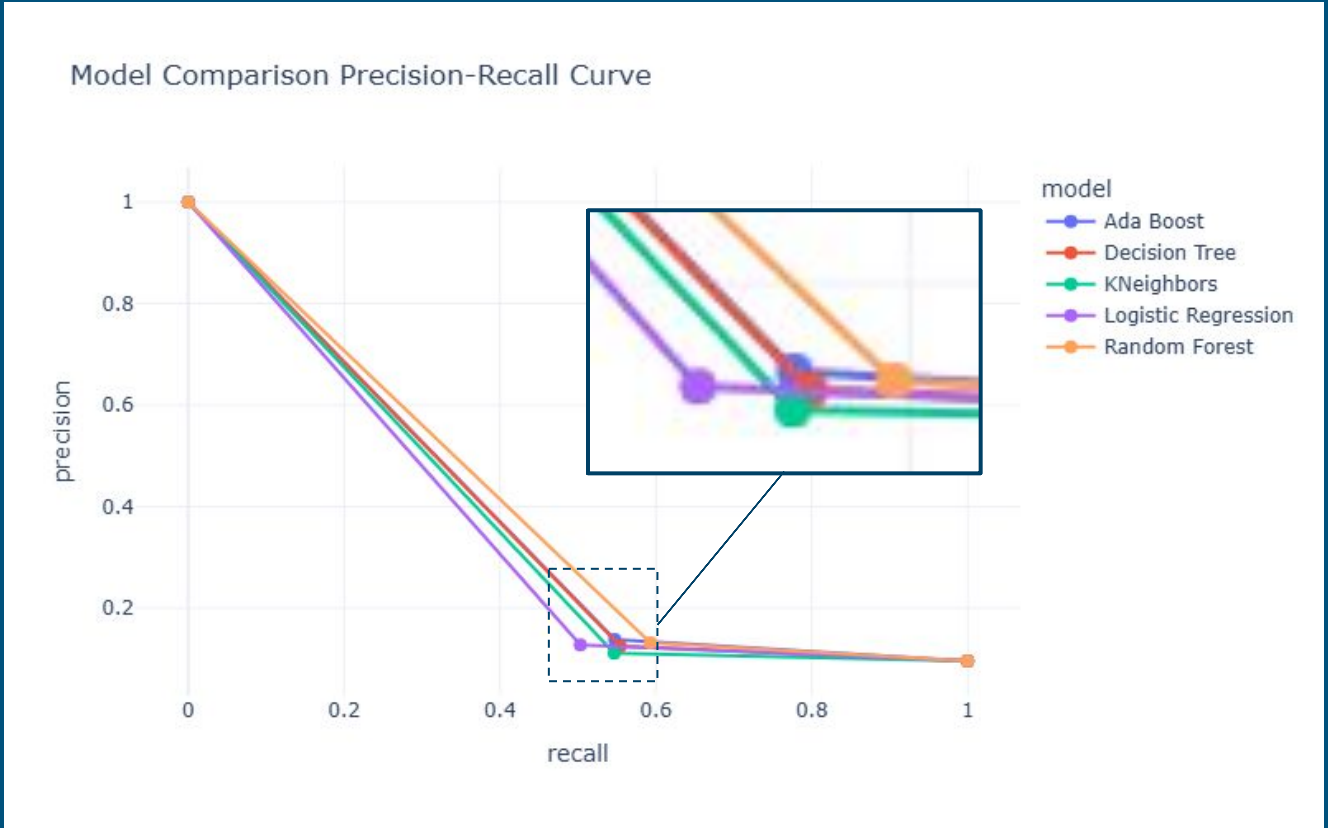
1. Logistic Regression
   - Assumes hospital encounters are independent of one another
     1. Duplicate patient numbers
   - Used for Feature importance
2. K-Nearest Neighbors
   - Assumes similar outcomes among similar patients
   - Requires feature selection
3. Decision Tree
   - Capture nonlinear relationships
4. Random Forest
   - Reduced overfitting compared to decision tree
5. AdaBoost
   - Well-suited for imbalance data

Hyperparameter Tuning Experiments

Experiment 1: Model Selection

- Goal: maximize recall
- 5-Fold Grid Search CV
- Pipeline Configuration
  - Random Under Sampler
  - Variance Threshold (0.3)
- Best
  - Random Forest
  - n_estimators: 50, max_depth: 10, max_features: None





Best recall: Random Forest (0.593)
Best precision: AdaBoost (0.138)

Experiment 2: Sampling Method

- Oversampler
  - Higher accuracy
  - Little improvement to other metrics
- Undersampler
  - Higher recall
  - Lower mean fit time (7 seconds vs 60 seconds)
- SMOTE
  - Best accuracy
  - Worst recall

Experiment 3: Feature Selection

- Select K Best
  - Selected top 13 features in feature importance + 2 feature engineered
  - Worst F1-score
- Variance Threshold (0.3)
  - Highest recall
- LinearSVC
  - Highest F1-score
  - Worst recall
  - Extremely Small improvement from variance (0.002)

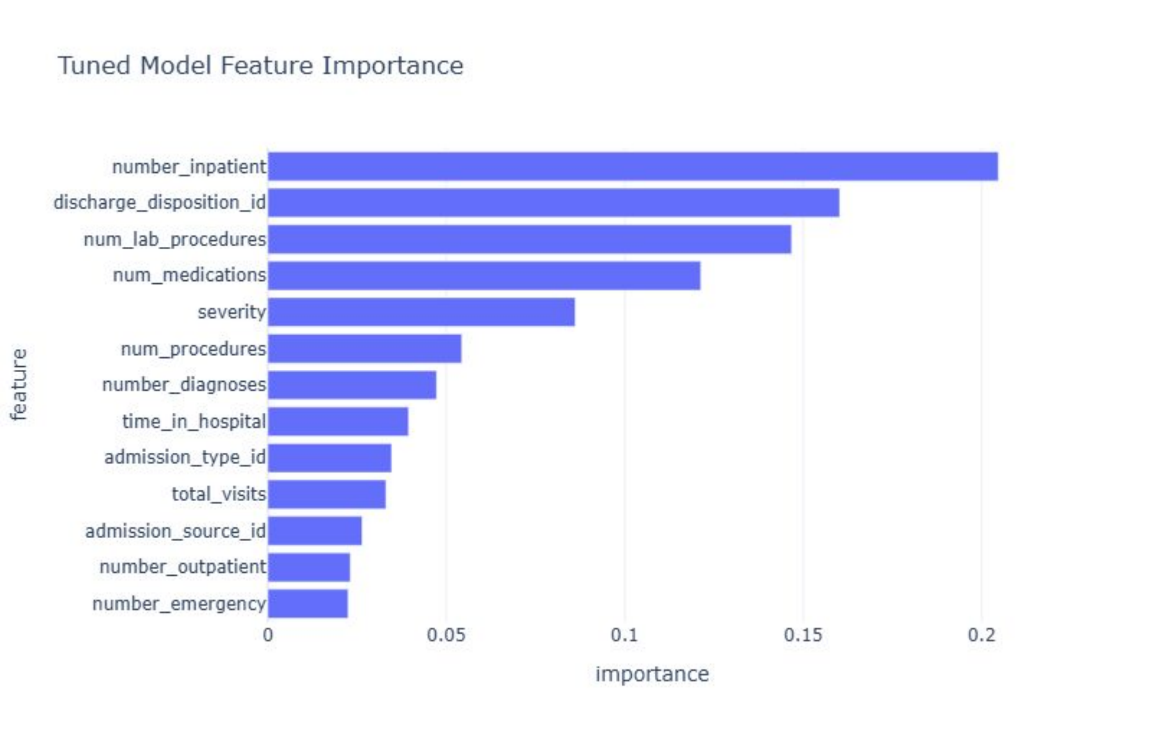
Final Metrics

Accuracy: 0.611

Precision: 0.166

Recall: 0.618

F1: 0.262



Limitations

- Data Limitations
  - Medication/medical_specialty feature imbalances
  - Missing data
    - A1C Result (83.3%)
    - Serum Glucose (94.7%)
    - Weight (96.9%)
  - Outdated (1999-2008)
- Model Limitations
  - One-hot encoding can create sparse data
  - May not capture complex relationships

Future Works

- Implement Neural Network (Deep Learning) Models
  - Potentially increase performance
- Validate Model With More Recent Data
  - Include patient physical characteristics/blood tests
  - Changes in medications may reduce side effects
- Add Weight To Medication Features

Workload Distribution

Tim

- Preprocessing
- EDA
- Modeling

Matt

- EDA
- Modeling
- Hyperparameter tuning

Any Questions?

Works Cited

Beata Strack, Jonathan P. DeShazo, Chris Gennings, Juan L. Olmo, Sebastian Ventura, Krzysztof J. Cios, and John N. Clore, “Impact of HbA1c Measurement on Hospital Readmission Rates: Analysis of 70,000 Clinical Database Patient Records,” BioMed Research International, vol. 2014, Article ID 781670, 11 pages, 2014.

Appendix: Feature Selection Hyperparameters

- Manual Selection
  - features = [
  - 'readmitted', # target
  - 'repaglinide',
  - 'rosiglitazone',
  - 'payer_code',
  - 'chlorpropamide',
  - 'age',
  - 'diag_2g',
  - 'specialty_grouped',
  - 'diag_1g',
  - 'pioglitazone',
  - 'number_inpatient',
  - 'severity',
  - 'total_visits']

- Variance Threshold
  - 0.3
- LinearSVC
  - C= 0.01
  - max_iter=10000
  - Threshold = median

Appendix: Tuning Experiment Parameters

- Logistic Regression
  - l1_ratio: [0, 0.5, 1.0]
  - C: [0.01, 0.1, 1.0, 10.0]
  - Solver: [‘saga’]
- K Neighbors Classifier
  - N_neighbors: [3, 5, 7]
  - Weights: [‘uniform’, ‘distance’]
- Decision Tree
  - Splitter: [‘best’, ‘random’]
  - Max_depth: [None, 10]
- Random Forest
  - N_estimators: [50, 100]
  - Max_depth: [None, 10]
  - Max_features: [“sqrt’, ‘log2’, None]
- AdaBoost
  - N_estimators: [50, 100]
  - Learning_rate: [0.5, 1.0]